

PROJECT REPORT

PROJECT TITLE: Habit Tracker with Data Visualization

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COURSE: Python Programming (Flipped Course)

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GITHUB REPOSITORY: <https://github.com/vaishnavi26mib10070-pixel/habit-tracker>

1. Introduction

Building and keeping up with daily habits usually comes down to consistency. However, staying consistent is hard without a simple way to track progress. The **Habit Tracker with Data Visualization** is a lightweight command-line tool written in Python that helps users stay on track. It operates offline without requiring accounts or sign-ups, letting users log their daily activities, keep tabs on their active streaks, and visualize their overall progress through generated charts.

2. Problem Statement

Many people struggle to stick with new habits because basic tracking methods—like pen-and-paper logs or plain text lists—don't provide clear feedback, streak tracking, or visual progress updates. Manual logging quickly becomes tedious. On the other hand, modern habit-tracking apps are often bloated with unnecessary features, require cloud accounts, or rely on an active internet connection.

This project tackles that issue by providing a simple, local, terminal-based solution. It offers quick logging, automatic streak tracking, and visual graphs without any extra setup.

3. Requirements Specification

3.1 Functional Requirements

- **Habit Management:** Users can add, view, edit, archive, and delete habits while organizing them into custom categories.
- **Daily Logging & Validation:** Users can mark habits as completed for the day. The system checks dates to prevent logging the same habit twice on a single day.
- **Streak Tracking Engine:** Automatically calculates consecutive completion days for each active habit.
- **Data Visualization:** Generates and exports two Matplotlib charts to disk: `category_chart.png` (a pie chart breaking down habits by category) and `completions_chart.png` (a bar chart showing daily progress).
- **Data Persistence:** Automatically saves all user data locally in a `habits.json` file so progress carries over across sessions.

3.2 Non-Functional Requirements

- **Performance:** Operations like menu selections, saving data, and rendering charts run almost instantly (taking less than half a second).
- **Usability:** A straightforward terminal menu guides the user, keeping the learning curve low.
- **Error Handling:** Gracefully handles invalid menu choices, missing files, or attempts to double-log habits without crashing.
- **Maintainability:** Built with a modular structure that separates the command-line interface, storage handling, streak logic, and chart creation into distinct modules.

4. System Architecture & Project Structure

The project uses a clean three-tier architecture:

1. **Presentation Layer:** `main.py` handles user interaction and the main CLI menu loop.
2. **Logic Layer:** The `habit_tracker/` directory contains core logic modules (`habits.py`, `tracker.py`, `streaks.py`, and `charts.py`).
3. **Data Layer:** `habits.json` handles local persistent storage.

`habit-tracker/`

├─ `main.py`

├─ `habits.json`

├─ `README.md`

├─ `statement.md`

├─ `.gitignore`

└─ `habit_tracker/`

 ├─ `__init__.py`

 ├─ `storage.py` # loads and saves `habits.json`

 ├─ `habits.py` # handles CRUD operations

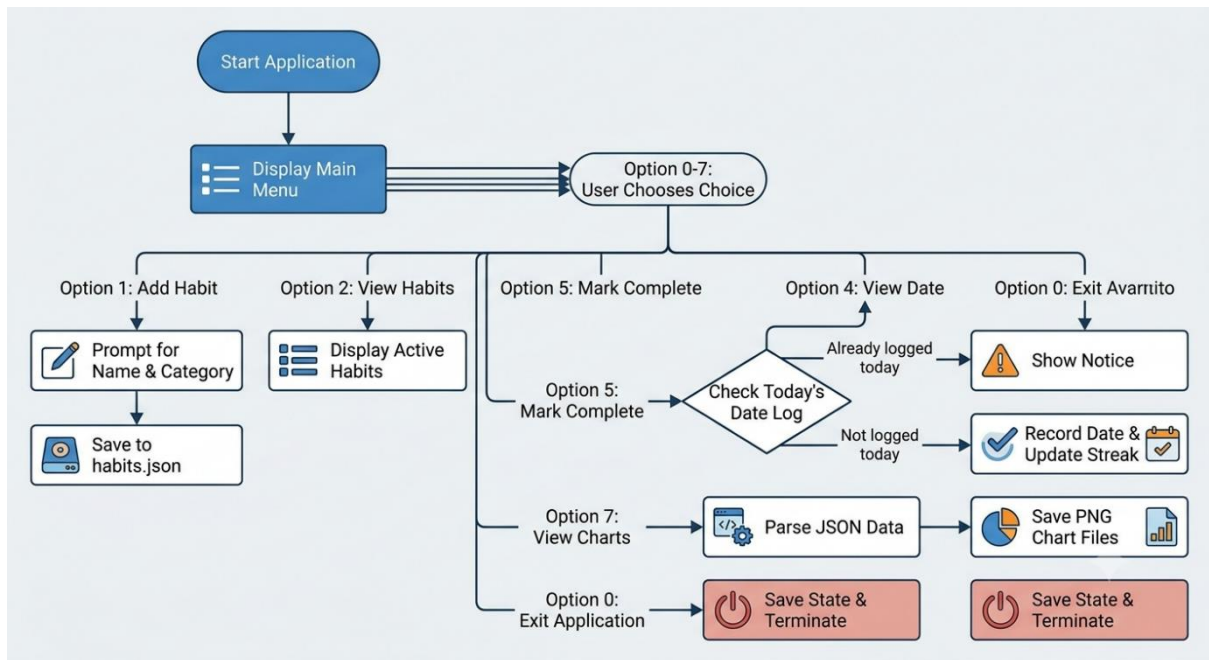
 ├─ `tracker.py` # handles daily completion marking

 ├─ `streaks.py` # calculates streak counts

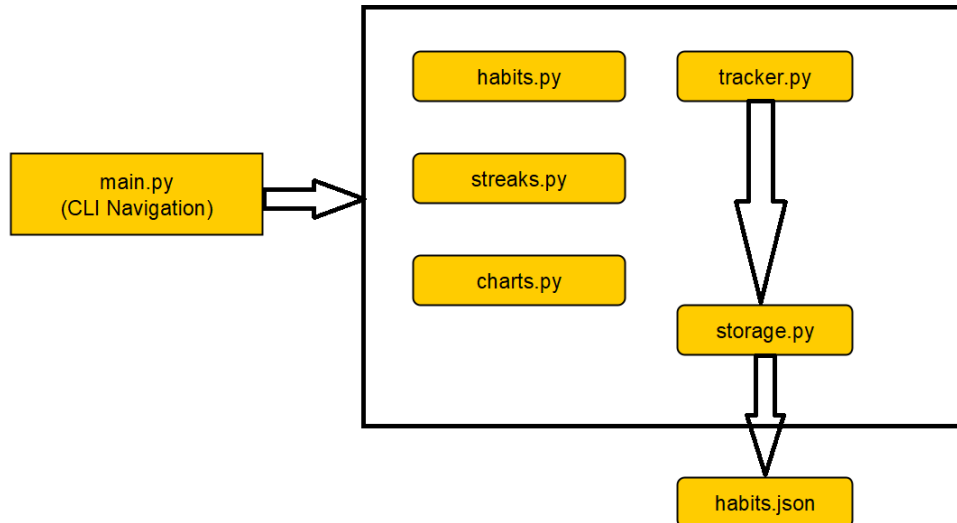
 └─ `charts.py` # generates pie and bar charts

5. Design Diagrams

5.1 Workflow Diagram



5.2 Component Diagram



5.3 Storage Schema (habits.json)

```
{
  "habits": [
    {
      "id": 1,
      "name": "Morning Workout",
      "category": "Health",
      "completion_dates": [
        "2026-03-28",
        "2026-03-29"
      ],
      "streak": 2
    }
  ]
}
```

6. Implementation Details & Key Features

- **Built With:** Written in Python 3 using standard libraries (json, datetime, os) alongside matplotlib for generating visual charts.
- **Duplicate Prevention:** Every time a user marks a habit complete, the tool checks if today's date formatted as YYYY-MM-DD already exists in that habit's completion_dates list. This keeps streak counts accurate and prevents duplicate logs.

7. Results & Visuals

(Insert screenshots from your project folder here in your final document)

- **Adding Habits & Navigation:** SS 1.png
- **Editing & Deleting Habits:** SS 2.png
- **Daily Completion Checks:** SS 3.png
- **Streak Calculations & Chart Triggers:** SS 4.png
- **Exported Visualizations:** category_chart.png and completions_chart.png

```
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit
Choose: 6
```

Habit Streaks

```
-----
Running: 1 day streak
Walking: 0 day streak
Talking with Samiksha: 1 day streak
Writing: 0 day streak
```

=== MY HABIT TRACKER ===

```
1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit
Choose: 7
Saved chart as category_chart.png
Saved chart as completions_chart.png
```

=== MY HABIT TRACKER ===

```
1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit
Choose: 0
Goodbye! Have a nice day!
```

=== MY HABIT TRACKER ===

1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit

Choose: 6

Habit Streaks

Running: 1 day streak
Walking: 0 day streak
Talking with Samiksha: 1 day streak
Writing: 0 day streak

=== MY HABIT TRACKER ===

1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit

Choose: 7

Saved chart as category_chart.png

Saved chart as completions_chart.png

```
ID | Habit Name | Category
-----
4 | Running | Health
5 | Walking | Health
7 | Talking with Samikshaaa | Fun
8 | Writing | Study
Enter the ID of the habit to edit: 7
New name (leave blank to keep 'Talking with Samikshaaa'): Talking with Samiksha
New category (leave blank to keep 'Fun'): FUN
Habit updated successfully!
```

```
=== MY HABIT TRACKER ===
1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit
Choose: 5
```

```
ID | Habit Name | Category
-----
4 | Running | Health
5 | Walking | Health
7 | Talking with Samiksha | FUN
8 | Writing | Study
Enter ID of the habit that you completed today: 7
Already marked as done for today.
```

=== MY HABIT TRACKER ===

1. Add New habit
 2. View All habits
 3. Delete a habit
 4. Edit a habit
 5. Mark habit as done today
 6. View streaks
 7. Show charts
 0. Exit
- Choose: 3

ID | Habit Name | Category

- | | | |
|---|-------------------------|----------|
| 1 | Study | Learning |
| 4 | Running | Health |
| 5 | Walking | Health |
| 7 | Talking with Samikshaaa | Fun |
| 8 | Writing | Study |

Enter the ID of the habit to delete: 1
'Study' has been deleted.

=== MY HABIT TRACKER ===

1. Add New habit
 2. View All habits
 3. Delete a habit
 4. Edit a habit
 5. Mark habit as done today
 6. View streaks
 7. Show charts
 0. Exit
- Choose: 4

ID | Habit Name | Category

- | | | |
|---|-------------------------|--------|
| 4 | Running | Health |
| 5 | Walking | Health |
| 7 | Talking with Samikshaaa | Fun |
| 8 | Writing | Study |

Enter the ID of the habit to edit: 7

```
PS C:\Users\Vaishnvi\OneDrive\Desktop\Habit tracker> python main.py
```

```
=== MY HABIT TRACKER ===
```

1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit

Choose: 1

Enter the habit name: Writing

Enter category (e.g. Health, Study): Study

Successfully Added 'Writing'!

```
=== MY HABIT TRACKER ===
```

1. Add New habit
2. View All habits
3. Delete a habit
4. Edit a habit
5. Mark habit as done today
6. View streaks
7. Show charts
0. Exit

Choose: 2

```
ID | Habit Name | Category
```

```
-----
```

1	Study	Learning
4	Running	Health
5	Walking	Health
7	Talking with Samikshaaa	Fun
8	Writing	Study

8. Testing Approach

- **Input Validation:** Tested menu choices using invalid input values, non-numeric strings, and out-of-range integers to verify that error messages handle exceptions properly.
- **Duplicate Protection:** Selected Option 5 multiple times on the same date to confirm that the app prevents duplicate completion logs.
- **Data Retention:** Added habits, logged streaks, closed the application, and re-opened it to confirm that the JSON storage retains state across sessions.

9. Challenges & Learnings

- **Challenges:** Setting up relative package imports inside a modular Python directory took some trial and error. Date handling for custom streak tracking logic also required careful parsing to handle edge cases accurately.
- **Key Takeaways:** Hands-on experience structuring modular Python code, manipulating JSON data structures, visualizing data with Matplotlib, and managing projects using Git version control.

10. Future Enhancements

- Building a desktop Graphical User Interface (GUI) using Tkinter or PyQt.
- Upgrading local storage from JSON to SQLite for long-term data tracking and analytics.
- Adding desktop notifications and reminders to nudge users when daily habits haven't been completed.

11. References

- [Python 3 Official Documentation](#)
- [Matplotlib Visualization Guide](#)
- [GitHub Documentation & Version Control](#)